

Data Visualization PROJECT REPORT
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Google Playstore Apps Analysis

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CERTIFICATE

This is to certify that LUTHFI NASSIR bearing Registration no 12223424 has completed INT2333 project titled, **“Google Playstore Apps Analysis”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Mrs Gargi Sharma

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Date: 18 / 11/ 2024

DECLARATION

I, Luthfi Nassir , student of B.Tech CSE under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 18/11/2024

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Luthfi Nassir

Acknowledgment

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Table of Contents

• Cover page	1
• Certificate	2
• Declaration	3
• Acknowledgement	4
• Introduction	6
• Scope of the Analysis	6
• Existing System	8
• Source of dataset	8
• ETL process	9
• Analysis on dataset	10
• List of Analysis with results	18
• Future scope	19
• References	19
• Bibliography	19

1. Introduction

The main objective of this report is to analyze a dataset obtained from Kaggle, which includes data on apps from the Google Play Store. By leveraging this data, this project aims to uncover key trends and insights into the mobile app industry. The dataset offers information on various attributes of mobile apps, such as their category, installation numbers, ratings, prices, and content ratings, all of which serve as the foundation for this analysis.

This report will utilize Tableau, a powerful data visualization tool, to create an intuitive and interactive dashboard that allows stakeholders to better understand the dynamics of the app market. Tableau is widely known for its ability to transform complex data sets into visually appealing and easy-to-understand graphics, making it ideal for the purpose of this analysis. The key goal is to provide insights that will help businesses, app developers, and marketers make more informed decisions regarding app development, marketing strategies, and market penetration.

The analysis will not only focus on the current trends in the app market but will also consider future possibilities, as the mobile application industry continues to grow at a rapid pace. The insights gained from this analysis could potentially inform decisions related to the development of new applications, marketing approaches, and understanding the behavior of app users across various regions.

In this report, we will examine the current state of the app market, explore key trends in app development and user behavior, and visualize these insights using various types of graphs and dashboards. By doing so, the report will provide a comprehensive understanding of the mobile app industry, backed by data-driven insights and actionable visualizations. Furthermore, the report will identify limitations in the current dataset and suggest possible areas for future research to gain an even deeper understanding of the dynamics shaping the mobile application market.

2. Scope of the Analysis

The scope of this analysis is focused on understanding the dynamics of the mobile application market using data sourced from the Google Play Store. The goal is to extract meaningful insights that can guide businesses, app developers, and marketers in making informed decisions regarding app development, marketing strategies, and market penetration. This analysis primarily focuses on the apps available in the Google Play Store and aims to uncover trends, patterns, and correlations that can offer a comprehensive view of the current state of the mobile app industry.

To achieve this, we will utilize a variety of techniques and methods that will allow us to explore several facets of the dataset, with each aspect contributing to a broader understanding of the mobile app landscape. The following areas will be the core focus of the analysis:

1. App Categories: The first area of analysis will explore the distribution of apps across different categories such as Games, Education, Finance, and Health & Fitness. This will provide a clear view of which categories dominate the app market, both in terms of the number of apps and their popularity among users. By analyzing the number of apps per category, we can identify trends such as saturation in specific categories and opportunities for new app development. Additionally, we will evaluate the correlation between app categories and user preferences to understand which categories have the highest demand.

2. Installation Trends: Understanding user behavior is crucial, and one of the best ways to do this is by examining the installation figures of apps. By analyzing the number of installations, we can assess which types of apps are performing well and which ones are underperforming. We will also look at how the number of installations correlates with factors such as app category, rating, and price. This analysis will offer insights into which types of apps attract the most users and which categories have the highest growth potential. Further, we will explore regional installation trends, highlighting which countries or regions have the highest concentration of users and apps.

3. App Ratings: App ratings are a critical metric in determining the success of an app. Higher ratings typically indicate better user satisfaction, while lower ratings may signal issues with the app. The analysis will involve examining the relationship between ratings and other app attributes such as category, installation count, and pricing. We will analyze whether certain categories have consistently higher ratings and whether this is linked to user engagement or the nature of the app itself. Additionally, we will explore whether apps with higher ratings tend to have more installations or whether a good rating directly impacts a user's likelihood to download an app.

4. App Pricing: Pricing analysis will focus on the distribution of paid vs. free apps across categories. We will explore how app pricing affects its performance, looking at whether higher-priced apps have fewer installs or if there is a correlation between price and rating. By evaluating the pricing structure of apps across categories, we can assess whether certain categories favor free apps over paid ones or vice versa. Additionally, we will analyze the maximum price of apps within each category to identify trends related to app pricing strategies.

5. Regional Analysis: Understanding geographical trends and user behavior across different regions is crucial for app developers and businesses looking to expand internationally. By analyzing the installation data from different regions, we can identify the most popular apps and app categories in various countries. This will provide valuable insights into regional preferences and can help developers tailor their marketing strategies for specific markets. Additionally, this analysis will offer insights into which regions are more inclined toward certain types of apps and how this influences the overall market strategy.

7. Insights for Future Development: The analysis will also touch upon potential future trends and areas of growth within the app market. By evaluating the current app categories, user behaviors, and ratings, we can predict where new opportunities might arise. For example, if we observe that certain app categories are rapidly growing in terms of installations, it may indicate

an emerging market that developers should target. Conversely, if a category is saturated, we might predict that growth opportunities lie in niche areas or in apps that cater to specific needs.

By conducting this analysis, we aim to not only present a snapshot of the current app market but also to offer insights that will be valuable for businesses, marketers, and developers looking to navigate the complexities of the mobile app ecosystem. The analysis will help identify trends, pinpoint areas for improvement, and provide a clearer understanding of how various factors influence the success of mobile applications.

3. Existing System

i. Drawbacks or Limitations of Existing System

The existing system relies on traditional business intelligence tools, which may not offer the flexibility or interactive capabilities that modern dashboard tools like Tableau provide. Some of the challenges with current systems include:

- **Data Inaccuracy:** The original dataset may contain inconsistencies or outdated information, such as text entries in numeric fields (e.g., “1M” or “100K”).
- **Manual Data Processing:** Some systems require manual adjustments for analyzing large datasets, which can be time-consuming.

4. Source of Dataset

The dataset used for this analysis is sourced from Kaggle, titled “**Google Playstore Apps**”. It contains detailed information about apps available on the Google Play Store. The dataset is available [here](#).

Overview of the Dataset

- **Number of Rows:** The dataset consists of **1,000,000 rows**.
- **Number of Columns:** There are **13 columns**, each providing different types of data for analysis.
- **Size of Dataset:** The dataset is with a size of about **633 MB**.
- **Data Types:**
 - **Categorical:** Columns such as Category, App Name, Type, Content Rating, and Genres are categorical, representing non-numeric values.
 - **Numerical:** Columns like Rating, Reviews, Size, Installs, and Price contain numerical values, though some require cleaning (e.g., "Installs" uses ranges like "1M" or "100K").

- **Text:** The App Name and Genres columns include text data, which may need further processing.

General Description of the Dataset

The dataset is structured with a variety of data types, offering insights into several aspects of the apps available on the Play Store. It includes the following key attributes:

- **App Name:** The name of the app.
- **Category:** The genre or type of app (e.g., Games, Education).
- **Rating:** The average rating given by users, though some rows have missing or inconsistent values.
- **Reviews:** The total number of reviews the app has received.
- **Size:** The size of the app in MB, but it may require conversion for consistency.
- **Installs:** The number of installations, listed in ranges (e.g., 10,000+), which requires cleaning.
- **Price:** The cost of the app, applicable only to paid apps.

5. ETL Process

The ETL process plays a critical role in preparing data for analysis, especially when dealing with datasets like the **Google Playstore Apps** dataset, which often requires cleaning and transformation before it can be used effectively in Tableau. The goal of the ETL process is to extract the data from its raw form, transform it into a structured format suitable for analysis, and then load it into Tableau for visualization.

Extracting the Data

The dataset is initially downloaded from Kaggle, where it is provided in CSV format. The raw data contains several issues, such as inconsistent formats and missing values, that must be addressed before any meaningful analysis can occur. For extraction purposes, this step is straightforward since the data is already in a tabular format that can be imported directly into Tableau or Tableau Prep.

Transforming the Data

Once the data is extracted, the transformation process begins. This is where most of the data cleaning and manipulation occur to make the data suitable for analysis in Tableau.

1. Standardizing 'Installs' Column:

- The **Installs** column contains values like '10,000+', '100,000+'. This format can cause issues for numerical analysis, as these values are not directly usable.

- In Tableau Prep, this can be done using calculated fields where you can use conditional statements to replace 'K' with *1000 and 'M' with *1000000, ensuring all entries are in numeric format.
- 2. **Standardizing 'Size' Column:**
 - The **Size** column also presents challenges, with values like '8.5M', '25K', and 'Varies with device'. The goal here is to standardize the size values into a single unit (such as KB).
 - Again, Tableau Prep allows you to use custom calculated fields to perform these transformations.
- 3. **Converting Data Types:**
 - Ensure that columns like **Price**, **Reviews**, and **Installs** are correctly classified as numerical fields. For example, the **Price** column might contain values like 'Free', which need to be converted to a numerical value (e.g., 0 for free apps).

Loading the Data

Once the data has been cleaned and transformed, it is ready to be loaded into Tableau for analysis. Tableau Prep allows for seamless export of cleaned data into Tableau, where it can be used for creating visualizations and dashboards. The cleaned dataset will now have consistent, numerical formats for installs, size, price, and rating, allowing for accurate and insightful visual analysis.

6. Analysis of Dataset

This section covers a detailed analysis of the **Google Playstore Apps** dataset, which includes several metrics that are crucial for understanding app performance, categories, pricing trends, and user ratings. Each analysis will present the graph created using Tableau and discuss its relevance, insights, and how it can be interpreted.

6.1 Top Categories by Number of Apps

Introduction

This analysis focuses on the distribution of apps across different categories in the Google Play Store. The **Top Categories by Number of Apps** graph is a bar chart that represents the number of apps available in each category.

General Description

In the Google Play Store, apps are divided into different categories such as **Games**, **Business**, **Health & Fitness**, **Education**, and more. The number of apps within each category can provide insights into the market saturation of each category, the diversity of offerings, and trends in app development.

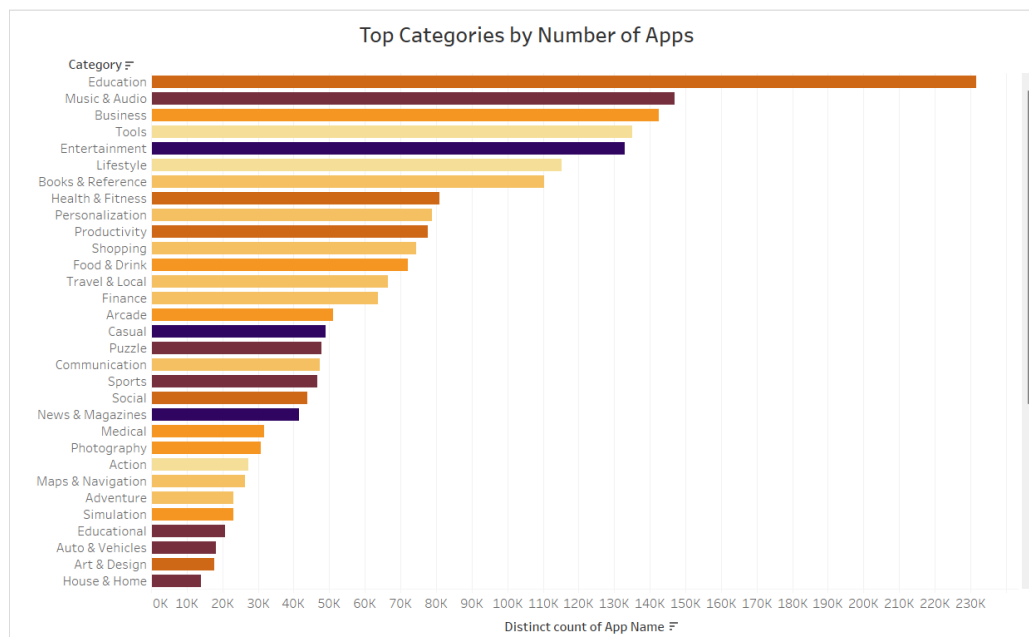
Specific Requirements, Functions, and Formulas

For this graph, we used the **Category** column from the dataset to count the number of apps in each category. This required grouping the data by category and counting the occurrences of apps in each group.

In Tableau:

1. Drag the **Category** field to the Rows shelf.
2. Drag **App Name** or **Number of Records** to the Columns shelf and change it to a count.
3. Apply sorting to display the categories with the highest number of apps.

Visualization



The bar chart clearly shows the comparison of app numbers across various categories, providing insights into which sectors are saturated and which are potentially underdeveloped.

6.2 Installation Across Categories (Treemap)

Introduction

The **Installation Across Categories** graph is a Treemap that shows the total number of installs for apps in each category. It provides a visual representation of how well different categories are performing in terms of user engagement.

General Description

Each app in the dataset has an associated number of installs, and this analysis aggregates the installs by category. The Treemap visually represents the proportional contribution of each category's total installs, where larger rectangles indicate categories with more installs.

Specific Requirements, Functions, and Formulas

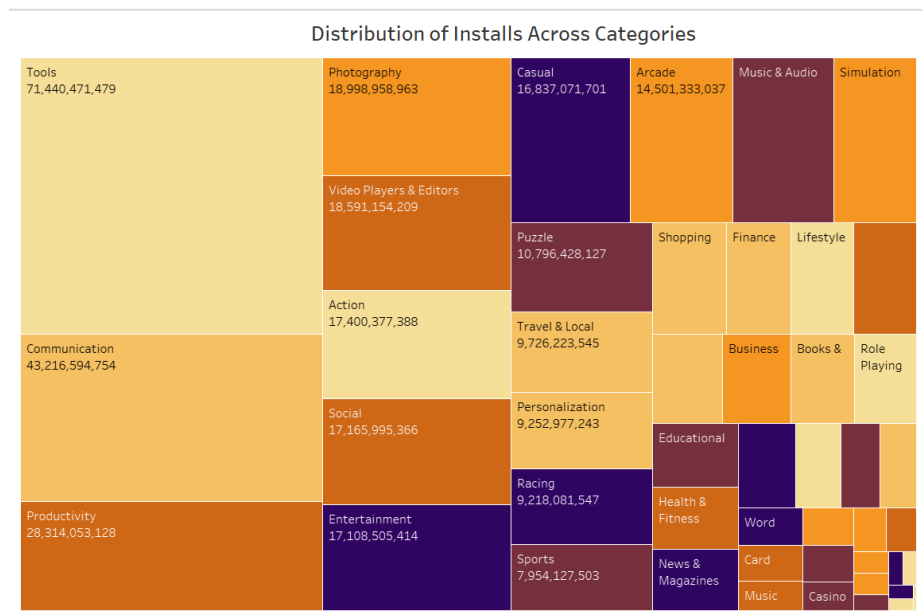
- The **Installs** column is used to sum the total number of installs for each category.
- To create the Treemap:
 1. Drag **Category** to the Rows shelf.
 2. Drag **Installs** to the Size shelf.
 3. Use the **Treemap** chart type in Tableau.

Analysis Results

This visualization allows us to understand which categories dominate in terms of user engagement:

- Categories like **Games** may show significantly higher install numbers compared to **Books** or **Education**.
- A category with a high install count but fewer apps might indicate high-performing individual apps.

Visualization



The Treemap provides a clear visual comparison of the relative popularity of each category, giving a quick overview of market dominance by installs.

6.3 Distribution of Apps by Rating (Bar Chart)

Introduction

The **Distribution of Apps by Rating** graph is a bar chart that illustrates how the apps are distributed across various rating levels. It shows the frequency of apps based on their user ratings, typically ranging from 1 to 5 stars.

General Description

The **Rating** column provides the average rating of apps based on user reviews. This analysis aims to understand the spread of ratings across all apps and identify trends in user satisfaction.

Specific Requirements, Functions, and Formulas

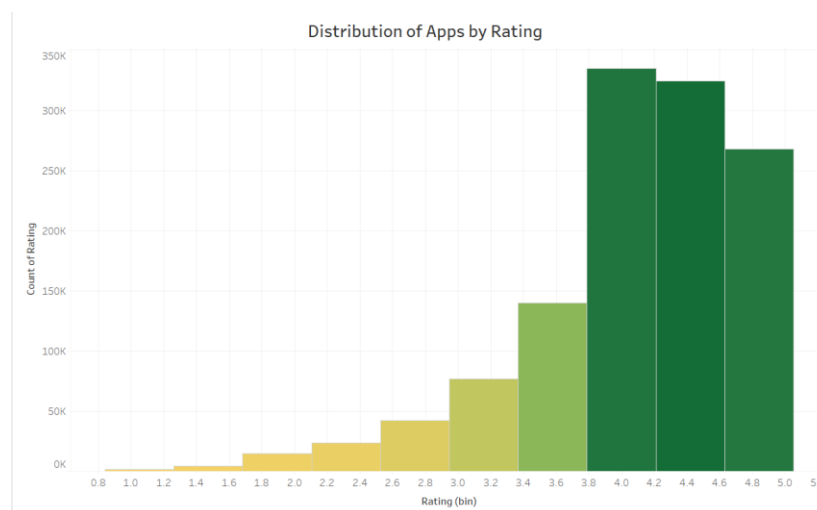
- Use the **Rating** column to group apps by their rating.
- The number of apps in each rating group is counted and visualized.
 1. Drag **Rating** to the Rows shelf.
 2. Drag **App Name** to the Columns shelf and count it.

Analysis Results

This graph shows how most apps tend to cluster around mid-range ratings (3 to 4 stars), indicating that users tend to rate apps relatively moderately.

- Apps with ratings of 1 or 5 stars are typically outliers, possibly due to extreme user experiences (either very positive or very negative).
- The distribution may suggest areas for improvement in app quality or customer service for apps with lower ratings.

Visualization



The bar chart clearly shows the frequency distribution of apps by their ratings, providing insight into user satisfaction trends across different apps.

6.4 Maximum Price in Each Category (Bar Chart)

Introduction

This analysis explores the **Maximum Price in Each Category** of apps. The bar chart displays the highest price of apps within each category, which can help identify pricing trends and pricing extremes across different app categories.

General Description

While many apps on the Google Play Store are free, some are paid apps, and their prices vary greatly. By looking at the maximum price in each category, we can identify which categories have the highest-priced apps and which categories might have more affordable offerings.

Specific Requirements, Functions, and Formulas

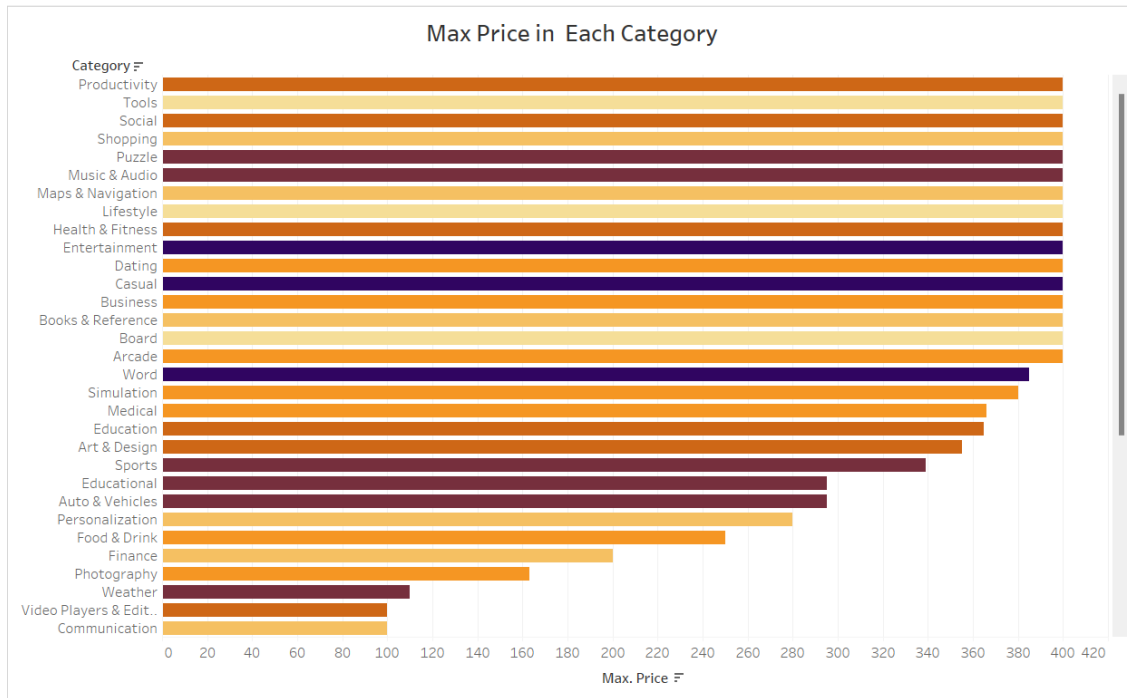
- Use the **Price** column and find the maximum price for each category.
- To create the graph:
 1. Drag **Category** to the Rows shelf.
 2. Drag **Price** to the Columns shelf and change it to "Maximum".

Analysis Results

This bar chart can reveal patterns in pricing:

- Categories like **Finance** or **Business** might have higher-priced apps due to premium services.
- **Games** and **Entertainment** categories may show a wider spread, with both high-priced premium apps and free apps.

Visualization



The bar chart allows for easy comparison of the maximum prices within each category, highlighting pricing trends.

6.5 KPI Metrics (Key Performance Indicators)

Introduction

KPIs provide a high-level view of the most important aspects of the dataset, including the average rating, average price, average number of installs, and the number of paid apps.

General Description

In this section, we have calculated KPIs such as:

- **Average Rating:** The average user rating across all apps.
- **Revenue:** The average price of apps.
- **Total Installs:** The average number of installs across all apps.

Specific Requirements, Functions, and Formulas

To calculate KPIs:

1. Use Tableau's built-in aggregation functions (AVG, COUNT, etc.) to calculate the values.

2. Create calculated fields for derived metrics such as percentage.

Analysis Results

KPIs provide a snapshot of the dataset:

- The **Average Rating** reveals the general quality of apps.
- The **Revenue** helps identify how much in each category the apps made.
- **Total Installs** indicates user engagement.
- **Percentage of Paid Apps** provides insights into how many apps are monetized.

Visualization

Total Apps	Average Rating	Total Installs	Total Revenue
2,177,712	2.203	424,278,878,315	1,461,176,544

KPIs can be displayed using **number cards** or **text tables** for a clean, easy-to-read presentation.

6.6 Revenue by Category (Bubble Graph)

Introduction

The **Revenue by Category** graph is a bubble chart that visualizes the total revenue generated by apps in each category. This analysis provides insights into the earning potential of different categories on the Google Play Store.

General Description

Revenue is calculated as the product of the price and the number of installs for each app. By aggregating this data by category, the bubble chart offers a proportional representation of how much revenue each category generates. The size of the bubbles represents the total revenue, making it easy to compare categories visually.

Specific Requirements, Functions, and Formulas

To calculate revenue:

- Use the formula:
Revenue = Price × Installs

Steps to create the graph in Tableau:

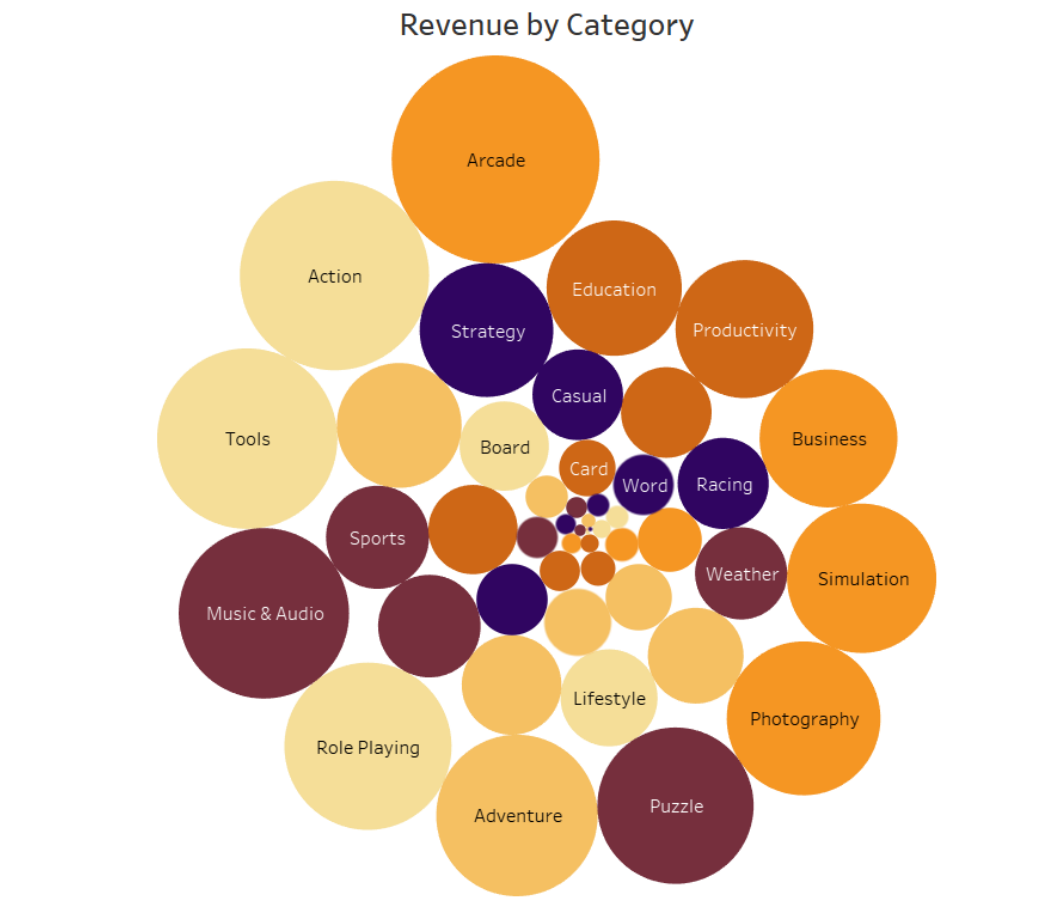
1. Create a calculated field for **Revenue** with the formula: Price * Installs.
2. Drag **Category** to the Rows shelf.
3. Drag **Revenue** to the Size shelf to control bubble sizes.
4. Drag **Revenue** to the Columns shelf and **Category** to the Rows shelf.
5. Use the **Bubble Chart** visualization type and adjust the sizes to make the chart visually appealing.

Analysis Results

This graph highlights:

- Categories like **Arcades** may dominate due to a high volume of installs and paid apps.
- Categories such as **Business** or **Productivity** may also show significant revenue due to higher-priced apps and specialized services.
- Categories with low revenue despite a high number of apps might indicate areas with untapped monetization opportunities.

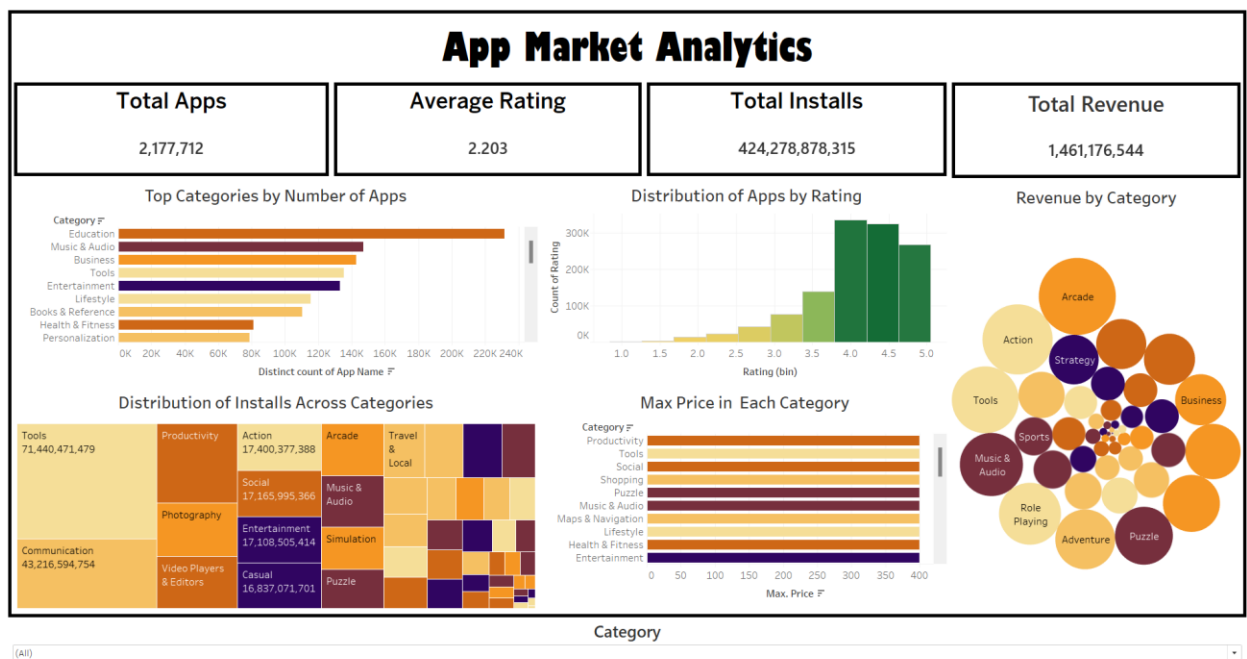
Visualization



The bubble chart provides an engaging and visually distinct way to represent revenue across categories. Larger bubbles signify categories with the highest earnings, while smaller bubbles indicate lower revenue generation.

7. List of Analysis with Results

- **Top Categories by Number of Apps:** A bar chart showing which app categories dominate the market in terms of the number of apps.
- **Installation Across Categories:** A Treemap showing the distribution of installs across categories.
- **Distribution of Apps by Rating:** A bar chart representing the frequency distribution of app ratings.
- **Maximum Price in Each Category:** A bar chart displaying the highest-priced apps in each category.
- **Revenue by Category:** A bubble chart showing the total revenue generated by apps in each category.
- **KPIs:** Summary of important metrics such as average ratings, prices, installs, and percentage of paid apps.



8. Future Scope

- **Extended Data Analysis:** Future work can explore additional dimensions, such as user demographics, revenue per app, and app updates.
- **Predictive Analytics:** Using machine learning models to predict app success based on attributes like price, category, and rating.
- **Integration with Real-Time Data:** Updating the dashboard with real-time data from the Playstore for more current insights.

9. References

- Kaggle Dataset: [Google Playstore Apps](#)
- Tableau Official Documentation: <https://www.tableau.com/learn>

10. Bibliography

1. **Google Play Store Apps Dataset**
Source: Kaggle
Link: [Google Play Store Apps Dataset](#)
Description: This dataset provided the foundation for the analysis. It includes app attributes such as category, installs, ratings, size, and pricing, used for generating insights and creating visualizations.
2. **Tableau Documentation**
Source: Tableau
Description: Comprehensive guide for creating visualizations, cleaning data, and building dashboards in Tableau.
3. **Data Preparation in Tableau Prep**
Source: Tableau Prep Blog
Description: Provided instructions and best practices for cleaning and preparing data for analysis using Tableau Prep.